9.8 Checkpoint Review

Date: 4/5/2026, 9:08:05 PM

Time Spent: 09:29

Score: 75%

Passing Score: 80%

Question 1

Correct

For Gigabit Ethernet and better, why is the performance of connectors increasingly critical?

- ○ Because Gigabit Ethernet uses more power.
- ◉ Because it uses all four pairs for transmission. ✓ Correct
- ○ Because it requires fiber optic connectors.
- ○ Because it only works with wireless connections.

Explanation

Gigabit Ethernet and higher categories use all four pairs of wires in the cable for transmission, which increases the sensitivity to crosstalk between the wire pairs. This makes the performance of connectors, which must minimize interference and maintain signal integrity, increasingly critical.

Power usage is not directly related to the critical nature of connector performance for Gigabit Ethernet.

Fiber optic connectors are used with fiber optic cables, not copper-based Ethernet cables.

Gigabit Ethernet and better are wired technologies, so their performance is not related to wireless connections.

Related Content

- 2.2.1 Unshielded Twisted Pair Cable
- 2.2.2 Shielded and Screened Twisted Pair Cable
- 2.4.2 Single Mode Fiber and Multimode Fiber
- 2.6.3 Cable Category Issues
- 2.6.8 Fiber Optic Cable Testing Tools

resources\questions\q_cable_category_four_pair_transmission_n09.question.xml

Question 2 ✕ Incorrect

What is the primary benefit of using a redundancy protocol like the Common Address Redundancy Protocol (CARP) in a high availability cluster?

- (●) It reduces the overall cost of the cluster. ✕ Incorrect
- ( ) It allows for unlimited scalability of the cluster.
- ( ) It enables failover to a passive node if the active one fails. ✓ Correct
- ( ) It increases the processing power of each node in the cluster.

Explanation

The primary benefit of using a redundancy protocol like CARP in a high availability cluster is to enable failover to a passive node if the active one fails. This ensures continuous service availability by automatically transitioning the virtual IP and workload to a standby node.

Redundancy protocols do not primarily aim to reduce costs; they focus on ensuring service availability through failover mechanisms.

While redundancy protocols support scalability to some extent, their primary benefit is not unlimited scalability but ensuring service availability.

The purpose of redundancy protocols is not to increase the processing power of individual nodes, but to manage failover and service continuity.

Related Content

resources\questions\q_clusters_redundancy_benefit_n09.question.xml

Question 3 Correct

A network engineer is tasked with optimizing the network performance of a Linux-based server. The engineer suspects that there are unnecessary services running that could be consuming network resources.

The network engineer wants to identify all services that are currently listening for incoming connections so they can review and potentially disable any that are not required.

Which **netstat** command should the network engineer use to identify these services?

- (selected) netstat -l ✓ Correct
- netstat -at
- netstat -s
- netstat -p

Explanation

The -l switch with netstat lists only ports in the listening state, which is exactly what the network engineer needs to identify services that are waiting for incoming connections. This command will help the engineer pinpoint potential candidates for optimization by revealing services that may not be necessary for the server's operation.

The -at switch combination shows all TCP connections in a listening or established state, which provides more information than the network engineer needs for identifying just the listening services.

The -s switch reports per-protocol statistics, which would not directly help the network engineer identify listening services.

On Linux, the -p switch shows the PID and process name associated with each connection, which could be useful for further investigation after identifying listening services, but it does not specifically filter for listening ports.

Related Content

- 1.3.7 Binary and Hexadecimal
- 2.3.5 Termination Tools and Techniques
- 4.1.3 Address Resolution Protocol
- 4.4.1 ipconfig
- 4.4.2 ifconfig and ip
- 4.4.3 arp

- 4.4.4 ping
- 4.4.5 Lab: IPv4 Troubleshooting Tools
- 4.4.6 Lab: IPv4 Troubleshooting tools for Linux
- 4.4.7 Lab: Use IPv4 Test Tools
- 4.6.1 IP Configuration Issues
- 4.6.2 Duplicate IP and MAC Address Issues
- 5.1.4 Packet Forwarding
- 5.1.7 Routing Table Tools
- 5.1.8 tracert and traceroute
- 5.1.10 Lab: Cisco Troubleshooting Tools
- 5.2.1 Dynamic Routing Protocols
- 5.3.2 Network Address Translation Types
- 5.3.3 Port Address Translation
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- 5.7.1 Routing Table Issues
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- 6.2.3 DHCP Options
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- 6.4.9 Lab: Troubleshoot IP Configuration 1
- 6.4.10 Lab: Troubleshoot IP Configuration 2
- 6.4.11 Lab: Troubleshoot IP Configuration 3
- 6.5.5 Host Address and Canonical Name Records

- 6.5.15 Lab: Troubleshoot DNS Records
- 6.5.16 Configuring DNS Caching on Linux
- 6.6.1 Client DNS Issues
- 6.6.2 Name Resolution Issues
- 6.6.3 nslookup
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- 6.6.5 Lab: Explore nslookup
- 6.6.6 Lab: Use nslookup
- 7.1.1 Transport Layer Security
- 8.2.2 Nmap
- 8.5.2 tcpdump
- 8.6.1 Common Performance Issues
- 8.6.6 Traffic Shaping
- 9.2.2 Attack Types
- 9.3.1 On-Path Attacks
- 9.3.6 VLAN Hopping Attacks
- 9.4.4 DNS Attacks
- 10.1.5 Digital Certificates and PKI
- 10.3.5 Lab: Scan for Unsecure Protocols
- 10.4.5 Extensible Authentication Protocol and IEEE 802.1X
- 10.4.6 Port Guards
- 12.1.2 IEEE 802.11a and 5GHz Channel Bandwidth
- 12.1.3 IEEE 802.11b/g and 2.4GHz Channel Bandwidth
- 12.2.1 Infrastructure Network Type
- 12.4.3 Channel Overlap Issues
- 13.2.4 Internet Key Exchange
- 13.3.9 Lab: Use PowerShell Remote

Question 4

Correct

What does a router configured as a default gateway use to determine the best path for forwarding packets?

- ○ Its MAC address table
- ◉ Its routing table ✓ Correct
- ○ Its ARP cache
- ○ Its DHCP leases

Explanation

The routing table contains information about the paths to various network destinations, allowing the router to make informed decisions about where to forward packets.

The MAC address table is used in switching to forward frames within the same network, not in routing packets to different networks.

The ARP cache maps IP addresses to MAC addresses for local network communication, not for determining routes to other networks.

DHCP leases are related to the assignment of IP addresses within the network, not the routing of packets.

Related Content

resources\questions\q_default_gate_forward_best_path_n09.question.xml

Question 5 Correct

What does TCP use to divide data from the Application layer?

- ○ Frames
- ○ Packets
- ◉ Segments ✓ Correct
- ○ Bits

Explanation

TCP divides the continuous stream of bytes received from the Application Layer into manageable pieces called segments. Each segment is then given a header that includes control information.

Frames are used in the Data Link Layer to encapsulate packets for transmission over physical media. TCP operates at the Transport Layer and uses segments to encapsulate data.

Packets are the primary unit of data used in the Network Layer. TCP divides application data into segments, which are encapsulated within IP packets for transmission.

Bits are the smallest unit of digital data but do not specifically refer to the method TCP uses to divide application data. TCP segments are composed of many bits but are a higher-level concept.

Related Content

- 4.1.1 IPv4 Datagram Header
- 4.1.3 Address Resolution Protocol
- 6.1.1 Transport Layer Ports and Connections
- 6.1.2 Transmission Control Protocol
- 6.1.3 TCP Handshake and Teardown
- 6.1.7 Lab: Explore Three-Way Handshake in Wireshark
- 7.1.1 Transport Layer Security
- 9.2.2 Attack Types

resources\questions\q_tcp_segments_n09.question.xml

Question 6 Correct

Why might downgrading software or firmware be necessary?

- To improve the system's security
- To fix a problem caused by a recent update ✓ Correct
- To increase the system's processing speed
- To reduce the cost of IT maintenance

Explanation

The correct answer is to fix a problem caused by a recent update. Downgrading may be required if a recent update or upgrade introduces issues or instability to the system, necessitating a return to a previous, stable version.

Downgrading typically involves reverting to an older version, which might not have the latest security patches.

Downgrading does not inherently improve processing speed; it's about system stability.

The decision to downgrade is based on functionality and stability, not maintenance costs.

Related Content

8.1.6 Lifecycle Management

8.1.12 Lab: Update Firmware

resources\questions\q_life_cycle_recent_update_problem_n09.question.xml

Question 7

Correct

Which of the following BEST describes the primary function of a SOHO router?

- A device that exclusively manages internal storage solutions for small offices and home offices
- A device that primarily forwards traffic between the Local Area Network (LAN) and the Wide Area Network (WAN), while also providing additional functionalities ✓ Correct
- A device designed solely for enhancing the security of personal devices within a home network
- A device that increases the range of wireless signals for devices connected to a Metropolitan Area Network (MAN)

Explanation

The primary function of a SOHO router is to forward traffic between the LAN and the WAN. This includes connecting devices within a small office or home office to the Internet (WAN) and facilitating communication between devices on the local network (LAN). In addition to routing, SOHO routers often provide additional functionalities such as firewall protection, wireless connectivity, and sometimes even VPN support.

Managing internal storage solutions is not the primary function of a SOHO router. This description more closely aligns with the role of a Network-Attached Storage (NAS) device, which provides centralized data storage and file sharing within a network.

While enhancing network security is an important function of many SOHO routers, it is not their sole purpose. SOHO routers are multifunctional devices that provide connectivity and network management features in addition to security measures.

The primary role of a SOHO router is not to increase the range of wireless signals for MAN connections. SOHO routers are designed for smaller networks, typically within a single building or home office, and their main function is to manage traffic between these local networks and the wider Internet. Extending wireless signal range, especially in the context of a MAN, would require different or additional networking equipment.

Related Content

resources\questions\q_soho_router_definition_n09.question.xml

Question 8 Correct

A network engineer is tasked with securing the transmission of data between the company's main office and its remote branch. The engineer needs to ensure that the data cannot be intercepted or tampered with during transmission.

Which of the following solutions should the network engineer implement to achieve this goal?

- Use cryptographic hash algorithms to hash all data before transmission.
- Implement TLS encryption for the data being transmitted. ✓ Correct
- Apply database encryption to the data before sending it over the network.
- Encrypt the data using a symmetric key algorithm and send the key along with the data.

Explanation

The correct answer is to implement TLS encryption for the data being transmitted. TLS (Transport Layer Security) encryption is specifically designed to secure data in transit, making it the optimal choice for protecting data transmitted over a network. It ensures that the data cannot be intercepted or tampered with during transmission by encrypting the data before it is sent and decrypting it upon receipt.

While cryptographic hash algorithms can verify the integrity of data, they do not encrypt data or protect it from interception during transmission.

Database encryption is intended to protect data at rest stored within a database, not data in transit.

Sending the encryption key along with the data over the same channel compromises the security of the transmission. If an attacker intercepts the data, they also obtain the key, rendering the encryption ineffective.

Related Content

- 1.3.5 Transport and Application Layer and Security Functions
- 5.2.5 Border Gateway Protocol
- 6.1.2 Transmission Control Protocol
- 9.1.4 Encryption
- 9.5.2 Password Attacks
- 10.1.5 Digital Certificates and PKI

resources\questions\q_encryption_tls_encryption_scenario_n09.question.xml

Question 9 ✕ Incorrect

Which type of connectors do SFP transceivers use?

- (●) RJ-45 ✕ Incorrect
- ( ) LC ✓ Correct
- ( ) SC
- ( ) MPO

Explanation

Small Form Factor Pluggable (SFP) transceivers use LC connectors, which are designed for fiber optic cables. LC connectors are known for their compact size and are commonly used in telecommunications and data communications for their excellent performance in connecting SFP transceivers to the network. This makes LC connectors the correct choice for SFP transceivers due to their reliability and efficiency in handling high-speed data transmissions.

RJ-45 connectors are typically used with twisted pair cables for Ethernet networking, not with fiber optic cables. While some transceivers, like certain types of SFP transceivers, can support RJ-45 connectors for copper networking applications, the standard and most common connector type for SFP transceivers, especially when referring to fiber optics, is the LC connector. Therefore, RJ-45 is not the correct answer for the type of connectors SFP transceivers use.

SC connectors are another type of fiber optic connector, known for their push-pull latching mechanism and larger size compared to LC connectors. While SC connectors are used in various fiber optic applications, SFP transceivers typically use LC connectors due to their smaller form factor and suitability for high-density applications. SC connectors are not the standard choice for SFP transceivers, making this option incorrect.

Multi-fiber Push On (MPO) connectors are designed for multi-fiber connections and are typically used in high-density fiber optic applications, such as data centers. MPO connectors can connect multiple fibers at once, supporting high bandwidth applications. However, SFP transceivers are designed for single or dual fiber connections using LC connectors, not the multi-fiber connections facilitated by MPO connectors. Therefore, MPO is not the correct answer for the type of connectors used by SFP transceivers.

Related Content

3.1.2 Modular Transceivers

resources\questions\q_modular_transceiver_connectors_n09.question.xml

Question 10 Correct

Which of the following ensures that power is supplied to a server or device during short power outages?

- (●) Uninterruptible power supply ✓ Correct
- ( ) Surge protector
- ( ) Line conditioner
- ( ) Backup generator

Explanation

An uninterruptible power supply (UPS) provides continuous power using batteries for a short period of time. Often, it is paired with a backup generator that can provide power over a longer time period.

Although a UPS often contains both surge protection and line conditioning, neither can maintain power during an outage.

Related Content

- 2.5.3 Power Management
- 3.4.1 Hardware Failure Issues
- 7.4.4 Fault Tolerance and Redundancy

resources\questions\q_power_ups_01_n09.question.xml

Question 11 Correct

What type of cabling do most Ethernet adapters designed for use with copper cabling now support?

- Coaxial cable
- Gigabit Ethernet ✓ Correct
- HDMI cable
- USB cable

Explanation

Most Ethernet adapters designed for use with copper cabling now support Gigabit Ethernet, which allows for high-speed data transmission over copper cables such as twisted pair cables.

Coaxial cable is used for different types of network and video connections but is not the primary type of cabling supported by modern Ethernet adapters for copper cabling.

HDMI cable is used for transmitting high-definition video and audio signals between devices and is not used for Ethernet network connections.

USB cable is used for connecting various peripherals to a computer but is not the type of cabling Ethernet adapters use for network connectivity.

Related Content

- 2.1.2 Ethernet Standards
- 2.1.3 Media Access Control and Collision Domains
- 2.1.4 100BASE-TX Fast Ethernet Standards
- 2.1.5 Gigabit Ethernet Standards
- 2.1.6 Fiber Ethernet Standards
- 2.1.7 Lab: Reconnect to an Ethernet Network
- 2.2.9 Lab: Connect to an Ethernet Network
- 2.7.1 Ethernet Standards
- 3.1.1 Network Interface Cards
- 3.1.2 Modular Transceivers
- 3.1.5 Ethernet Frame Format

14.1.4 Fibre Channel

14.3.3 Cloud Gateways

resources\questions\q_nic_gigabit_ethernet_n09.question.xml

Question 12 ✕ Incorrect

What is the default probe message type used by the **traceroute** command?

- (●) ICMP Echo Request ✕ Incorrect
- ( ) TCP SYN
- ( ) UDP ✓ Correct
- ( ) ARP Request

Explanation

The **traceroute** command uses UDP probe messages by default to trace the path between two nodes. This method helps in identifying the route and diagnosing any issues along the path.

ICMP Echo Request is incorrect because this is the default probe type used by the tracert command on Windows systems, not traceroute.

TCP SYN is incorrect as traceroute does not use TCP SYN packets by default for its probes.

ARP Request is incorrect because ARP Requests are used for resolving IP addresses to MAC addresses within a local network, not for tracing the path between two nodes over the Internet or a large network.

Related Content

- 1.3.7 Binary and Hexadecimal
- 2.3.5 Termination Tools and Techniques
- 4.1.3 Address Resolution Protocol
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- 6.6.3 nslookup
- 6.6.4 dig

- 6.6.5 Lab: Explore nslookup
- 6.6.6 Lab: Use nslookup
- 7.1.1 Transport Layer Security
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- 12.4.3 Channel Overlap Issues
- 13.2.4 Internet Key Exchange
- 13.3.9 Lab: Use PowerShell Remote

resources\questions\q_tracert_udp_message_type_n09.question.xml

Question 13 Correct

You are setting up a new office network and need to configure a Windows server to use a static IP address to ensure it always has the same address for network services.

The IP address you have been allocated for the server is 192.168.1.100, with a subnet mask of 255.255.255.0, and the default gateway for the network is 192.168.1.1.

Which of the following netsh commands should you use to configure the server's Ethernet adapter?

- ○ netsh interface ip set address "Ethernet" dhcp
- ◉ netsh interface ip set address "Ethernet" static 192.168.1.100 255.255.255.0 192.168.1.1 ✓ Correct
- ○ netsh interface ip renew "Ethernet"
- ○ netsh interface ip show config "Ethernet"

Explanation

The correct command for this scenario is **netsh interface ip set address "Ethernet" static 192.168.1.100 255.255.255.0 192.168.1.1**. This command correctly uses the **netsh** command to configure the Ethernet adapter with a static IP address (192.168.1.100), subnet mask (255.255.255.0), and default gateway (192.168.1.1), which matches the requirements given you for your network setup.

The **netsh interface ip set address "Ethernet" dhcp** command configures the adapter to obtain an IP address automatically via DHCP, which is not suitable for a server that needs a static IP.

The **netsh interface ip renew "Ethernet"** command is used to renew the DHCP lease for an adapter, not to set a static IP address.

The **netsh interface ip show config "Ethernet"** command displays the current configuration and does not set or change the IP configuration.

Related Content

resources\questions\q_win_config_netsh_interface_scenario_01_n09.question.xml

Question 14 Correct

What does the **show config** command display on a switch?

- ○ The switch's IP address
- ◉ The switch's current configuration ✓ Correct
- ○ The list of connected devices
- ○ The switch's version information

Explanation

The **show config** command is used to display the current configuration of a switch. This is crucial for administrators to understand the current settings and operations of the switch.

The switch's IP address is typically shown in the **show ip interface brief** command, not **show config**.

The list of connected devices can be viewed with commands like **show mac address-table**, not **show config**.

The switch's version information is typically shown in the **show version** command, not **show config**.

Related Content

3.4.3 Switch Show Commands

3.4.8 Lab: Troubleshoot Disabled Ports

resources\questions\q_int_config_show_config_n09.question.xml

Question 15 Correct

Which of the following BEST describes the major infrastructure of the Internet?

- A network of personal computers connected via Bluetooth
- High bandwidth trunks connecting Internet eXchange Points (IXPs) ✓ Correct
- A series of underground tunnels used exclusively for fiber-optic cables
- Satellite constellations providing global wireless access

Explanation

The backbone of the Internet consists of high-capacity data transmission lines, known as trunks, which connect various Internet Exchange Points around the world. IXPs are key components where different networks, including those of ISPs, connect and exchange traffic, facilitating the global reach of the Internet.

Bluetooth is a short-range wireless technology primarily used for connecting devices over short distances, such as between a computer and a mouse or keyboard. It does not form the major infrastructure of the Internet, which requires long-distance, high-capacity connections.

While fiber-optic cables are a significant part of the Internet's infrastructure, providing high-speed data transmission, they are not exclusively housed in underground tunnels. The infrastructure of the Internet is more complex and includes above-ground and undersea cables, satellite links, and various types of networking equipment, not just underground tunnels.

Satellite constellations do contribute to global Internet access, especially in remote areas where traditional infrastructure is not feasible. However, they do not represent the major infrastructure of the Internet. The core infrastructure relies more heavily on terrestrial networks and undersea cables connecting IXPs for the bulk of data transmission.

Related Content

- 1.2.1 Open Systems Interconnection Model
- 1.2.5 Layer 3 - Network
- 1.2.8 OSI Model Summary
- 1.3.4 Network Layer Functions
- 1.3.6 The Internet
- 1.3.7 Binary and Hexadecimal
- 1.3.8 Lab: Explore a Single Location in a Lab

- 4.1.2 Layer 2 vs Layer 3 Addressing and Forwarding
- 4.2.3 Subnet Masks
- 6.2.2 DHCP Server Configuration
- 13.1.1 Wide Area Networks and the OSI Model
- 14.3.5 Cloud Firewall Security

resources\questions\q_internet_ixp_n09.question.xml

Question 16 Correct

What is the difference between stateless and stateful modes in DHCPv6?

- Stateless mode provides a routable IP address, while stateful mode does not.
- Stateless mode uses multicast for discovery, while stateful mode uses broadcast.
- Stateful mode provides a routable IP address and additional options, while stateless mode only provides additional options. ✓ Correct
- Stateful mode is used exclusively for IPv4 networks, while stateless mode is used for IPv6 networks.

Explanation

In stateful mode, a host can obtain a routable IP address from a DHCPv6 scope, in addition to any other options, similar to DHCP for IPv4. In contrast, stateless mode is primarily used to provide additional option settings without offering leases for host IP addresses.

Stateless mode provides a routable IP address, while stateful mode does not is incorrect because it reverses the roles of stateless and stateful modes.

Both stateless and stateful modes use multicast for discovery, as IPv6 does not support broadcast.

Both stateless and stateful modes are used in IPv6 networks, not IPv4.

Related Content

- 4.1.2 Layer 2 vs Layer 3 Addressing and Forwarding
- 5.7.3 VLAN Assignment Issues
- 6.2.1 DHCP Process
- 6.2.2 DHCP Server Configuration
- 6.2.3 DHCP Options
- 6.2.4 DHCP Reservations and Exclusions
- 6.2.5 Lab: Configure a DHCP Server
- 6.2.6 Lab: Configure DHCP Server Options
- 6.2.7 Lab: Create DHCP Exclusions
- 6.2.8 Lab: Create DHCP Client Reservations

- 6.2.9 Configure Client Addressing
- 6.2.10 Lab: Configure Client Addressing for DHCP
- 6.3.2 IPv6 Interface Autoconfiguration and Testing
- 6.3.3 DHCPv6 Server Configuration
- 6.3.6 Set Up Alternate Addressing
- 6.4.1 DHCP Relay and IP Helper
- 6.4.2 DHCP Issues
- 6.4.3 Troubleshooting DHCP Exhaustion
- 6.4.4 Lab: Configure a DHCP Relay Agent
- 6.4.5 Lab: Add a DHCP Server on Another Subnet
- 6.4.6 Lab: Troubleshoot Address Pool Exhaustion
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- 6.4.11 Lab: Troubleshoot IP Configuration 3
- 6.6.1 Client DNS Issues

resources\questions\q_dhcpv6_stateless_vs_stateful_n09.question.xml

Question 17 Correct

A server is running Microsoft SQL Server and is replicating the data to other Microsoft SQL servers on the network.

Which principle port is the application service using?

- (selected) TCP 1433 ✓ Correct
- TCP 1521
- TCP 3306
- TCP 5432

Explanation

Microsoft SQL Server uses TCP/1433 to allow clients to connect to the database server over the network and allow replication traffic to move between database servers. Microsoft SQL Server uses TCP/1433.

Oracle's remote data access protocol SQL*Net uses TCP/1521 to allow clients to connect to the database server over the network and allows replication traffic to move between database servers.

The open-source MySQL platform and the MariaDB platform both use TCP/3306 to allow clients to connect to the database server over the network and allows replication traffic to move between database servers.

The open-source PostgreSQL platform uses TCP/5432 to allow clients to connect to the database server over the network and allows replication traffic to move between database servers.

Related Content

- 6.1.6 Common TCP and UDP Ports
- 7.2.7 Database Services
- 7.2.10 Lesson Review

resources\questions\q_db_services_tcp_1433_n09.question.xml

Question 18 ✕ Incorrect

Which switch can be used with **tracert** to specify a timeout in milliseconds?

- (●) -d ✕ Incorrect
- ( ) -h
- ( ) -w ✓ Correct
- ( ) -6

Explanation

The **-w** switch can be used with **tracert** to specify a timeout in milliseconds. This allows the user to adjust how long tracert waits for a response before considering a probe lost.

The **-d** switch is incorrect because this switch is used to suppress name resolution, not to specify a timeout.

The **-h** switch is incorrect as this switch is used to specify the maximum number of hops, not a timeout.

The **-6** switch is incorrect because this switch forces **tracert** to use IPv6 instead of IPv4, not to adjust the timeout.

Related Content

- 1.3.7 Binary and Hexadecimal
- 2.3.5 Termination Tools and Techniques
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- 6.4.9 Lab: Troubleshoot IP Configuration 1
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- 12.2.1 Infrastructure Network Type
- 12.4.3 Channel Overlap Issues
- 13.2.4 Internet Key Exchange
- 13.3.9 Lab: Use PowerShell Remote

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Question 19 Correct

What can cause a broadcast storm in a network?

- Low CPU utilization
- Insufficient storage space on network devices
- Switching loops ✓ Correct
- High bandwidth availability

Explanation

Switching loops occur when there are multiple paths between switches that allow data frames to circulate indefinitely. This can lead to a broadcast storm, where the network is flooded with redundant copies of the same data packets, consuming all available bandwidth and potentially causing network devices to fail. The Spanning Tree Protocol (STP) and similar technologies are designed to prevent such loops by disabling redundant paths, ensuring a loop-free topology.

Low CPU utilization is not a cause of broadcast storms. Broadcast storms are caused by structural issues in the network topology, not by the performance or utilization levels of devices.

Insufficient storage space on network devices would not lead to broadcast storms. Storage capacity issues might affect device functionality but would not cause the network-wide issues characteristic of broadcast storms.

High bandwidth availability does not cause broadcast storms. While having more bandwidth can mitigate the impact of increased traffic, it does not address the root cause of a broadcast storm, which is a topological issue.

Related Content

- 3.3.3 Spanning Tree Protocol
- 3.3.4 Spanning Tree Protocol Configuration
- 3.4.2 Port Status Indicators
- 3.4.3 Switch Show Commands
- 3.4.5 MAC Address Table
- 3.4.6 Network Loop and Broadcast Storm Issues
- 3.4.9 Lab: Switching Loop
- 8.2.5 Performance Monitoring
- 8.2.6 Availability Monitoring

9.3.4 MAC Flooding Attack

9.3.6 VLAN Hopping Attacks

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Question 20

Incorrect

What is a security control?

- A tool designed to eliminate all risks in an IT system.
- A procedure or tool designed to give a system properties. ✓ Correct
- A policy that restricts employee use of company IT resources. ✗ Incorrect
- A legal requirement for data protection.

Explanation

Security controls are procedures, activities, and tools designed to protect the confidentiality, integrity, availability, and non-repudiation of a system or data asset. They are part of an organization's security measures.

Security controls aim to mitigate risks, not eliminate all risks.

While some security controls may involve policies on the use of IT resources, this is too narrow a definition.

Legal requirements for data protection may necessitate security controls, but they are not security controls themselves.

Related Content

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